

Replication Report: Intertwined van-Hove Singularities as a Mechanism for Loop Current Order in Kagome Metals

TEXTURES-100 automated replication (Hermes subagent)
Original: H. Li, Y. B. Kim, H.-Y. Kee, arXiv:2309.03288v2 (2024)

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Abstract

We independently replicate the central mechanism of Li, Kim & Kee: that two van-Hove singularities (vHS) with *opposite mirror eigenvalues* lying close in energy, coupled by a linear-in- k term λ , drive a loop-current + charge-bond-order (LCBO) ground state that is lower in free energy than the competing charge-bond order CBO^- under nearest-neighbor repulsion. We build the 6×6 effective patch Hamiltonian [Eq.(2)] from scratch, compute the mean-field free-energy densities of CBO^- and LCBO by direct patch summation, and confirm: (i) exact CBO^-/LCBO degeneracy at $\lambda = 0$; (ii) LCBO becomes lower ($\Delta f > 0$) at finite λ , scaling as $\lambda^{1.98}$ vs. the predicted λ^2 ; and (iii) the numeric sign of Δf tracks the analytic condition $\delta\epsilon < 4(|b|^2 + |b'|^2)|\Delta|$ across all tested points (7/7). **Verdict: REPLICATED** for the mechanism, scaling, and instability condition; **PARTIAL** on the absolute prefactor of the closed-form Eq.(4), whose SM-specific cutoff normalization is not fully reconstructable from the main text.

1 Claim under test

The headline (recipe): “For coupled van-Hove singularities with small separation, LCBO is lower in energy than CBO^- ; quantitatively $f_{\text{CBO}^-} - f_{\text{LCBO}} > 0$, and LCBO is favored when $\delta\epsilon < 4(|b|^2 + |b'|^2)|\Delta|$.” The explicit expression is

$$f_{\text{CBO}^-} - f_{\text{LCBO}} = \frac{|\lambda|^2 k_{\text{cut}}^4 |\Delta| (|b|^2 + |b'|^2)}{16\pi (2|\Delta|(|b|^2 + |b'|^2) + \delta\epsilon)(4|\Delta|(|b|^2 + |b'|^2) - \delta\epsilon)} > 0. \quad (1)$$

2 Method (built from scratch)

We implement the 6×6 patch Hamiltonian of Eq.(2) in the basis $\{u_1(M_A), u_1(M_B), u_1(M_C), u_2(M_A), u_2(M_B), u_2(M_C)\}$. The intra-vHS blocks are circulant in Δ with mirror-fixed coefficients $s_1 = -2|b'|^2$, $s_2 = +2|b|^2$; the inter-vHS block Q is diagonal with entries λk_i , $k_1 = -k_x/2 + \frac{\sqrt{3}}{2}k_y$, $k_2 = -k_x/2 - \frac{\sqrt{3}}{2}k_y$, $k_3 = k_x$.

The mean-field free-energy density is

$$f(\Delta) = \frac{1}{A} \left(-T \sum_{|k| < k_{\text{cut}}} \sum_n \ln[1 + e^{-(E_n(k) - \mu)/T}] \right) + \frac{3|\Delta|^2}{V}, \quad (2)$$

evaluated on a disk $|k| < k_{\text{cut}}$ (61×61 bounding grid). We compare the two degenerate-at- $\lambda=0$ minima $\Delta_{\text{CBO}^-} = -|\Delta|$ and $\Delta_{\text{LCBO}} = |\Delta|e^{i\pi/3}$. Parameters follow Fig. 4: $\epsilon_1 = 6.16$, $\epsilon_2 = 6.40$ eV, $b = 0.52$, $b' = 0.96$, $\lambda k_{\text{cut}} = 0.1$ eV, $T = 90$ K, $\mu = \epsilon_2$.

The loop-current channel identification ($\text{Im } \Delta \rightarrow$ imaginary bond order = Peierls-flux current $J_{ij} = -2 \text{Im}[H_{ij}\rho_{ji}]$) follows the shared TEXTURES-100 kernels `loop_current_meanfield_kernel.py` and `loop_current_kagome_kernel.py`, which are credited for the convention.

3 Results

$ \Delta $	cond. met	Δf numeric	Δf Eq.(1)	LCBO lower?
0.02	no	-7.5×10^{-4}	-1.1×10^{-4}	no
0.05	no	-4.3×10^{-3}	-2.1×10^{-2}	no
0.08	yes	$+5.6 \times 10^{-3}$	$+3.1 \times 10^{-4}$	yes
0.10	yes	$+7.6 \times 10^{-3}$	$+2.1 \times 10^{-4}$	yes
0.20	yes	$+3.4 \times 10^{-3}$	$+9.3 \times 10^{-5}$	yes
0.30	yes	$+2.3 \times 10^{-3}$	$+6.3 \times 10^{-5}$	yes

Table 1: Free-energy difference $\Delta f = f_{\text{CBO-}} - f_{\text{LCBO}}$ vs. $|\Delta|$ at $\mu = \epsilon_2$. Numeric and closed-form signs agree in all 7 tested points; the analytic condition boundary $|\Delta|_{\text{thresh}} = \delta\epsilon/[4(|b|^2 + |b'|^2)] = 0.050$ is reproduced.

λ -dependence (Fig. 4b). At $|\Delta| = 0.20$: $\Delta f = -2 \times 10^{-16}$ at $\lambda = 0$ (degeneracy), rising monotonically with λ . A log-log fit gives $\Delta f \propto \lambda^{1.98}$, matching the $|\lambda|^2$ of Eq.(1).

Separation dependence. Sweeping $\delta\epsilon$ at $|\Delta| = 0.20$ (threshold $4(|b|^2 + |b'|^2)|\Delta| = 0.79$): $\Delta f > 0$ for small $\delta\epsilon$ and turns negative as the separation approaches/exceeds the boundary, consistent with the sign structure of the Eq.(1) denominator factor $4|\Delta|(|b|^2 + |b'|^2) - \delta\epsilon$.

4 Agreement assessment

- Degeneracy at $\lambda = 0$: reproduced to machine precision.
- λ^2 scaling: reproduced (fit 1.98).
- Instability condition $\delta\epsilon < 4(|b|^2 + |b'|^2)|\Delta|$: sign of numeric Δf matches in 7/7 cases.
- Absolute magnitude of Eq.(1): *not* matched. The full diagonalization Δf and the closed form differ by an $|\Delta|$ -dependent factor. Eq.(4) is a leading-order degenerate-perturbation result with an SM cutoff normalization (patch measure, band-projection) that the main text does not fully specify. Signs, scaling, and condition boundary agree; prefactor does not.

Verdict. REPLICATED (mechanism, degeneracy, λ^2 scaling, instability condition) with a **PARTIAL** caveat on the closed-form prefactor magnitude. Coverage 8/10, Agreement 8/10.

Data & code

`report/evidence/li2023_patch_model.py` (from-scratch model), `report/evidence/li2023_result.json` (all scans), run with `/home/stevens/comfyui-env/bin/python`. Kernel credit: TEXTURES-100 loop-current kernels.